

today. As 3GIO uses hot-plugging capabilities, devices can simply be added and removed without opening the PC chassis and without powering down the system first.

Intel is compatible with PCI at the software layers, so if the software supports PCI, it automatically supports 3GIO too. The initial speed of a single channel or 'lane', which is comprised of two differentially driven pair of wires (transmit and receive), is 2.5 Gbps in each direction. The lanes will supposedly scale to 10 gigabits per pin and beyond in the future.

The PCI bus is parallel in nature. The difference between serial and parallel buses from a motherboard designer's standpoint is the number of traces and routes needed between the device and core logic. Parallel architectures require a significantly higher amount of trace points than serial designs and, thus, the number of pins needed to correspond to these routes goes up as well. Changes such as these not only increase cost, but also cause a rise in signalling problems due to electromagnetic interference.

A new type of controller introduced in third generation I/O architecture is the switch. A switch essentially manages interconnections streams of data and between attached devices. In the case of the PCI bus, multiple devices share the same pathway for data flow. With Hyper-Transport and 3GIO, each device receives a dedicated link from the device to the core logic. This setup overcomes some of the problems faced by PCI such as system resource conflicts and inefficient use of bandwidth. Since devices reside on independent bus links, each link can be scalable to a higher width if the need arises.

A key advantage that 3GIO holds over HyperTransport is the focus on external devices. Not only does 3GIO act as a point-to-point link between chips, the standard is also explicitly designed to link to external devices. HyperTransport, on the other hand, is more focused on connecting the internal devices of a PC.

Current implementations of HyperTransport, such as nVidia's nForce, focus upon bus links between the north and south bridges. HyperTransport has no problem in performing as a system-wide bus.

3GIO, like HyperTransport, can handle bus connections across the system. Interestingly, the Arapahoe group has opted in favour of positioning 3GIO as more of an expansion bus in the immediate future. A more comprehensive system-wide approach may be taken in the future. However, at the moment, nothing has been etched in stone.

Announced at the spring session of the Intel Developer Forum, the Arapahoe architecture was recently handed over to the PCI SIG (Special Interest Group), the industry organisation

chartered to develop and manage the PCI standard, which will assume further governing and development of 3GIO as a standard in the future.

The last word

AMD and Intel are both aiming to solve the same problem with similar constraints. Therefore, it is not surprising for them to come up with similar solutions. Between these two solutions, HyperTransport appears to be a faster way to connect devices today. Since it uses parallel bus architecture, it should not be very difficult to quickly implement with existing technology. The biggest advantage that HyperTransport has over 3GIO is that it is available now. In fact, nVidia has already incorporated HyperTransport technology in their upcoming nForce chipset. Transmeta too has jumped onto the bandwagon and grabbed a license for the technology. Other proponents of this technology such as API Networks announced a chip design that would let computers use HyperTransport.

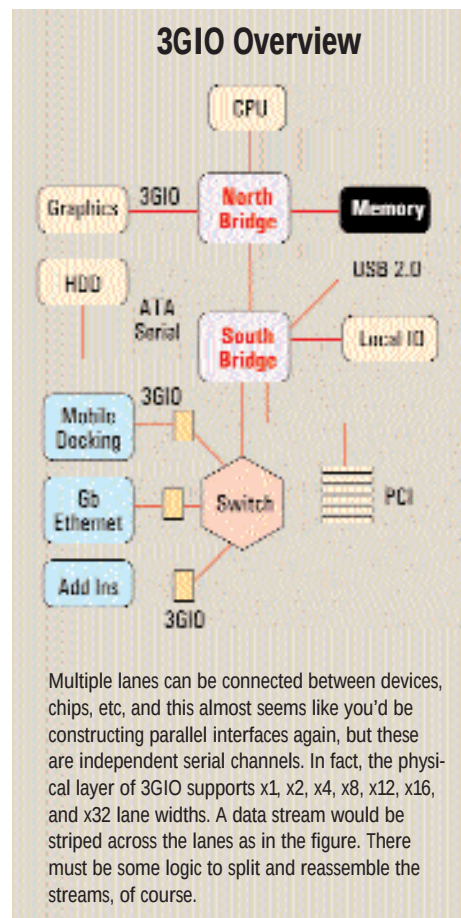
So far, 3GIO can only be considered 'on paper' since products supporting Intel's third generation I/O specification will not be seen until 2003.

But before the AMD camp starts cheering, here's a pause for thought: 3GIO has been adopted by the PCI SIG (Special Interest Group), the influential standards body which governs the currently prevailing PCI bus standards. Considering how widespread the PCI bus is and how comfortable companies are with the group and the PCI standard itself, its magic could easily rub off on 3GIO.

Surprisingly, nothing is holding back the two competing solutions to work together. With HyperTransport at the chip level and 3GIO linking external devices, a perfect blend of complementary technologies would result. This, however, will most likely not happen.

Two overlapping standards raise indecision for component makers, who must decide whether to support one standard, the other, or both.

In the end, Intel just might be a little too late in getting to the market with 3GIO. Technology solutions based upon HyperTransport have already been designed and implemented into silicon, some of which are very close to being released in the market. □



Multiple lanes can be connected between devices, chips, etc, and this almost seems like you'd be constructing parallel interfaces again, but these are independent serial channels. In fact, the physical layer of 3GIO supports x1, x2, x4, x8, x12, x16, and x32 lane widths. A data stream would be striped across the lanes as in the figure. There must be some logic to split and reassemble the streams, of course.